Exploring and Enhancing Placement of Intrusion Detection Systems in RPL: A Federated Learning-Based Approach

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Project Repository: https://github.com/mdilyas-27/cns_project_team_19

Abstract*—Routing protocol for low-power and lossy networks (RPL) forms the routing backbone for energy-constrained Internet of Things (IoT) deployments. Despite its efficiency, RPL is fundamentally susceptible to internal routing attacks executed by compromised nodes. While recent literature proposes diverse machine learning-based intrusion detection systems (IDSs), the strategic placement of monitoring engines within multi-hop RPL topologies remains largely uncharted. This study evaluates three architectural paradigms—Centralized with Local Information (CIDwL), Centralized with Global Information (CIDwG), and Distributed Collaborative ID (DCID)—across 31,626 Cooja simulation scenarios in a 10-level Destination-Oriented Directed Acyclic Graph (DODAG). We systematically investigate three foundational research questions regarding spatial detection degradation, optimal monitor density, and consensus communication overhead. Findings reveal that centralized root monitoring collapses on distant downstream attacks (detection accuracy dropping to 51.5%), whereas multi-hop global streaming induces severe communication overhead (378 kB). To reconcile this conflict, we evaluate FedID, a decentralized Federated Learning framework combining XGBoost with Federated Averaging (FedAvg). FedID eliminates runtime communication overhead entirely (0 bytes in testing), protects data privacy, and achieves up to 98.0% detection accuracy in RPL attacks alongside 99.60% accuracy on the benchmark CICIoT2023 dataset. Finally, we formulate a concrete novelty extension for upcoming iterations—Dynamic and Resilient FedID (DR-FedID)—incorporating entropy-driven dynamic monitor rotation, Multi-Krum Byzantine aggregation, and Local Differential Privacy.

***Index Terms*—RPL, Low-Power and Lossy Networks (LLNs), Intrusion Detection System (IDS), Placement Strategy, Federated Learning, XGBoost, Non-IID Traffic, Byzantine Resilience, IoT Security.**

I. INTRODUCTION

The Internet of Things (IoT) is increasingly deployed across mission-critical domains, ranging from industrial telemetry and intelligent transport systems to smart grids and medical cyber-physical infrastructure. Forecasts estimate that connected IoT edge devices will exceed 29 billion by 2030. These deployments fundamentally rely on Low-Power and Lossy Networks (LLNs), which are governed by extreme physical and operational constraints: 16-bit microcontrollers, strict memory bounds (under 10 KB of RAM), multi-year battery lifespans, and lossy, low-throughput wireless channels operating under IEEE 802.15.4 standards.

To manage multi-hop routing under such scarce resources, the IETF standardized the Routing Protocol for Low-Power and Lossy Networks (RPL, RFC 6550) [1]. RPL constructs a Destination-Oriented Directed Acyclic Graph (DODAG) rooted at a central border router (6LBR). Using objective functions such as MRHOF (based on the Expected Transmission Count, ETX) or OF0, RPL establishes energy-efficient, loop-free upward and downward forwarding topologies. However, RPL is intrinsically vulnerable to insider routing threats [2]. Once an authenticated node is compromised or captured, its legitimate cryptographic keys allow it to broadcast illegitimate routing states or drop forwarded traffic, severely disrupting network operations without violating link-layer security checks.

Intrusion Detection Systems (IDSs) provide the essential secondary line of defense against such insider threats. However, existing academic literature exhibits a severe *algorithm bias*: researchers focus overwhelmingly on proposing classifier architectures (SVM, Random Forests, Deep Neural Networks, LSTM) while assuming an idealized central observer with omniscient visibility into all network packets. The decisive question of **IDS Placement**—where monitoring agents should physically and logically reside within a multi-hop DODAG—has remained largely uninvestigated.

This project provides an exhaustive multi-criteria evaluation of IDS placement across centralized and distributed paradigms, analyzes the empirical findings of the FedID framework [3], and formulates a novel architectural extension for upcoming iterations.

II. RELATED WORK

Machine learning-based intrusion detection for IoT networks has received significant attention. Verma and Ranga [4] developed ensemble classifiers for RPL attacks, while Napiah et al. [5] analyzed 6LoWPAN compression headers. Shukla [6] evaluated wormhole detection using K-means and decision trees. However, these works evaluated models under an implicit centralized paradigm where all network telemetry is assumed to be instantaneously accessible at the root node.

Recent studies highlight the practical shortcomings of centralized collection in LLNs. Raza et al. [7] emphasized that continuous transmission of high-dimensional feature vectors to a central sink exhausts node batteries and saturates lossy 802.15.4 channels. Distributed trust-based models have been explored by Khan and Herrmann [8] and Medjek et al. [9], but these introduce heavy local computational burdens on resource-scarce motes.

Federated Learning (FL), pioneered by McMahan et al. [10], enables decentralized model training without raw data exchange. While FL has been applied to general IoT intrusion detection (e.g., Abbas et al. [11]), its integration with multi-hop RPL topologies and its sensitivity to topological Non-IID traffic distributions across DODAG hierarchy levels have not been comprehensively characterized prior to the work of Yilmaz et al. [3].

III. PROBLEM FORMULATION AND METHODOLOGY

A. Routing Protocol and Threat Formulation

RPL configures nodes into a DODAG via four ICMPv6 control messages: DODAG Information Object (DIO) for downward parameter advertisement and Rank broadcast; DODAG Information Solicitation (DIS) to solicit immediate DIOs; Destination Advertisement Object (DAO) sent upward to establish downward routes; and DAO-ACK. The scalar *Rank* dictates topological hierarchy, monotonically increasing downward away from the root.

We analyze seven classical RPL insider attacks executed across varied topological depths:

1) *Decreased Rank (DR):* Attacker broadcasts an illegally reduced Rank to hijack parent status and divert subtree traffic.
2) *Increased Version (IV):* Attacker illegitimately increments the DODAG version number, forcing costly global repairs.
3) *Blackhole (BH):* Attacker silently drops 100% of forwarded data packets.
4) *Selective Forwarding (SF):* Attacker probabilistically discards specific packets while forwarding others.

5) *DIS Flooding (DI):* Continual transmission of DIS probes repeatedly resets neighbor Trickle timers.
6) *DIO Flooding (HF):* High-frequency broadcast of bogus routing DIOs saturating wireless buffers.
7) *Wormhole (WP):* Colluding nodes establish an out-of-band tunnel to advertise artificial shortcuts.

B. Architectural Placement Models

We formalize three fundamental placement architectures:

1) **CIDwL (Central Local):** A standalone IDS engine deployed strictly at the root node (ID:0), relying solely on passively collected local traffic:

$$y_{pred}(t) = M_{root}(X_{root}(t)) \quad (1)$$

2) **CIDwG (Central Global):** The root maintains the classifier, but K collaborative nodes across the DODAG periodically forward 35-dimensional feature vectors to the root:

$$y_{pred}(t) = M_{root}([X_{root}(t), X_1(t), ..., X_K(t)]) \quad (2)$$

3) **DCID (Distributed Collaborative):** Each monitor k independently extracts features, performs local inference, and raises a local alarm $a_k \in \{0, 1\}$. Alarms are aggregated at the root under voting threshold θ:

$$y_{global} = \mathbf{I}((1/K) \sum_{k=1}^{K} a_k \geq \theta) \quad (3)$$

C. Federated Intrusion Detection (FedID)

FedID [3] replaces raw feature streaming with model parameter exchanges. Collaborators train local XGBoost tree ensembles on their local traffic. The root aggregates incoming tree parameters via Federated Averaging (FedAvg):

$$W_{global}(t+1) = \sum_{k=1}^{K} (n_k / N) W_k(t+1) \quad (4)$$

Crucially, during inference/testing, the global model executes **autonomously at the root node**, incurring zero packet transmissions ($O_{comm} = 0$).

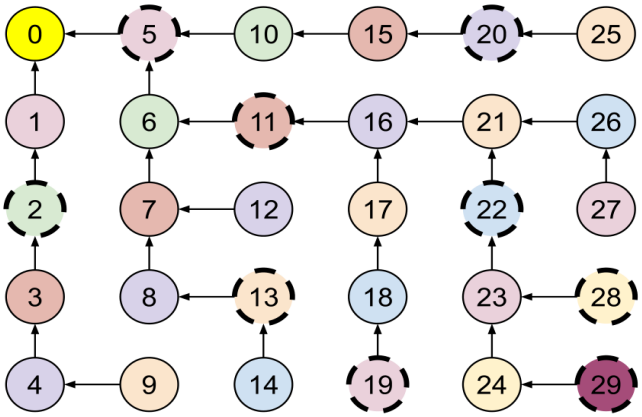


Fig. 1. Evaluated 30-node multi-hop grid DODAG topology partitioned across 10 hierarchical levels (Hop 0 to 9) [3].

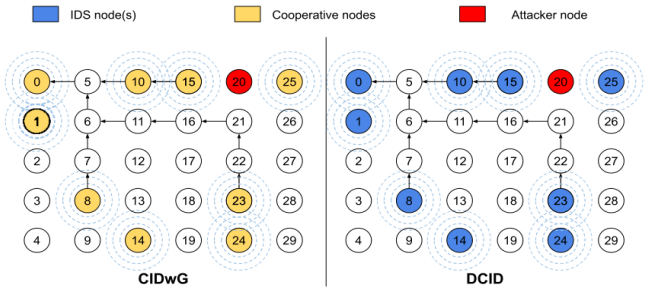


Fig. 2. Collaborator deployment across DODAG levels for CIDwG and DCID architectures [3].

IV. DATA AND EXPERIMENTAL PROTOCOL

A. Simulation Environment and Hardware

Simulations are executed in Cooja within Contiki OS 2.7, emulating Zolertia Z1 motes equipped with 16-bit MSP430 microcontrollers (8 KB RAM, 92 KB Flash) and CC2420 transceivers. As detailed in Table I, the deployment spans a 100 m x 80 m grid with 30 nodes arranged in 10 DODAG levels. Radio propagation uses the Unit Disk Graph Model (UDGM) with distance loss and a 25 m transmission range. Nodes transmit periodic UDP packets to the sink every 15 seconds over a 5-hour runtime per scenario across 31,626 distinct executions.

TABLE I
SIMULATION CONFIGURATION PARAMETERS [3]

Parameter	Experimental Value
Network OS	Contiki OS 2.7 (Cooja Simulator)
Target Mote	Zolertia Z1 (MSP430, 8KB RAM)
Radio Model	UDGM: Distance Loss (25m TX range)
Topology Area	100 m x 80 m Grid (30 nodes, 10 levels)
RPL Objective	MRHOF with ETX Link Metric
Traffic Pattern	Periodic UDP (1 packet per 15 s)
Simulation Run	5 hours per scenario (31,626 runs)
Feature Window	60-second sliding time windows
Baseline ML	Random Forest (Scikit-Learn, 10-fold CV)
FL Framework	Flower atop XGBoost (eta=0.3, depth=6)

B. Feature Extraction and Non-IID Nature

Traffic is discretized into 60-second non-overlapping sliding windows. From each window, 35 features are computed at the network and routing layers [12]: packet counts of DIO, DIS, and DAO control messages sent/received; data forwarding and drop ratios; rank churn and version instability metrics; and packet inter-arrival statistics. Because traffic density depends on topological depth (sub-root nodes aggregate traffic while leaf nodes remain mostly idle), local datasets are fundamentally Non-IID.

C. Benchmark Dataset (CICIoT2023)

To confirm generalizability beyond simulated RPL traces, models were evaluated on the CICIoT2023 real-time dataset [13], featuring 33 IoT attack types grouped into 7 classes (DDoS, DoS, Recon, Web, Brute Force, Spoofing, Mirai).

V. RESULTS

A. RQ1: Spatial Degradation in Centralized Monitoring (CIDwL)

In CIDwL, only the root node (ID:0) monitors traffic. Attacker nodes were systematically deployed at levels 1 through 9. As documented in Table II and Fig. 3, detection accuracy degrades severely as attacker distance increases:

• *Blackhole (BH):* Accuracy peaks at 98.8% when the attacker is at Level 1, but drops to 51.5% at Level 8 (a 47.3% drop). Discarded packets never reach the root.
• *Selective Forwarding (SF):* Achieves poor accuracy across all levels (maximum 58.0%).
• *DIS Flooding (DI):* Plummets from 99.2% (Level 2) to 49.7% (Level 8) due to localized Trickle reset effects.
• *Increased Version (IV):* Remains above 99.3% network-wide as version churn propagates globally.
Resolution of RQ1: A single central IDS is fundamentally insufficient for detecting localized or downstream attacks in RPL networks.

TABLE II
CIDWL DETECTION ACCURACY AT ROOT NODE (ID:0) [3]

Atk	L1	L3	L5	L7	L9	Max Drop
BH	0.988	0.533	0.530	0.540	0.562	0.473
SF	0.580	0.528	0.540	0.508	0.512	0.075
DI	0.967	0.557	0.520	0.570	0.503	0.495
DR	0.500	0.568	0.535	0.528	0.538	0.132
HF	1.000	0.802	0.582	0.592	0.553	0.483
IV	0.998	1.000	0.998	0.997	0.993	0.007
WP	0.927	0.517	0.510	0.522	0.493	0.433

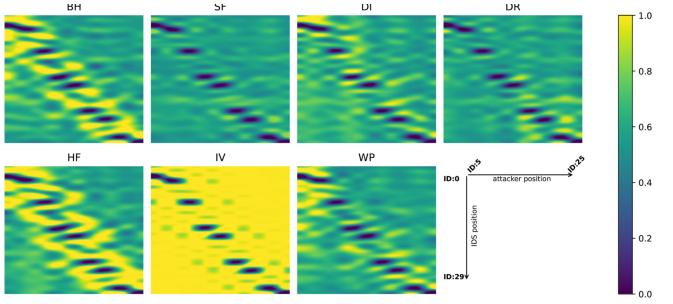


Fig. 3. Heatmap of detection accuracy vs. attacker position (x-axis) vs. IDS monitor position (y-axis) under CIDwL [3].

B. RQ2: Optimal Monitor Density Across Levels (CIDwG)

In CIDwG, distributed collaborators forward feature vectors to the root. As shown in Fig. 4, detection rises monotonically with monitor count: Blackhole detection improves from 78.3% (2 nodes) to 99.4% (9 nodes). True Positive Rate (TPR) increases by 21.0%, while False Positive Rate (FPR) drops from 22.2% to 0.3%. Hello Flooding reaches 100% TPR with 0.3% FPR under 9 collaborators.
Resolution of RQ2: Effective network-wide detection requires enlisting **at least one monitoring node per DODAG level**.

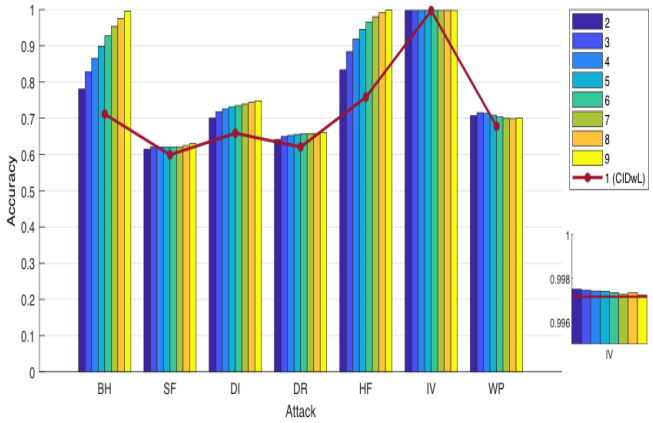


Fig. 4. Overall accuracy: CIDwL baseline vs. CIDwG with 2 to 9 distributed collaborators [3].

C. RQ3: Distributed Consensus Voting (DCID)

DCID delegates classification to local nodes and transmits lightweight alarm tokens to the root. Across five voting thresholds (Minority, 50% Majority, 60%, 70%, 80%), **50% Majority Voting** provides the optimal balance, achieving 80.5% (BH), 86.6% (HF), and 99.8% (IV) accuracy while lowering communication overhead from 378 kB to 2,700 bytes (Fig. 5).
Resolution of RQ3: A **50% majority voting consensus** minimizes message transmission while maintaining reliable collaborative detection.

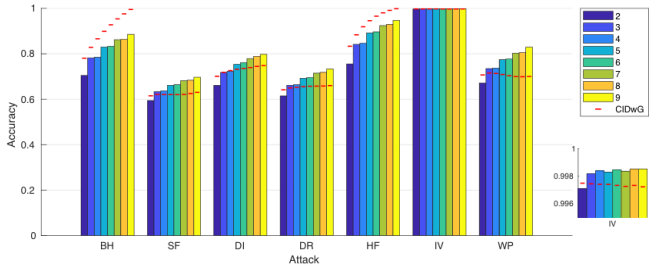


Fig. 5. DCID overall accuracy across attack types with 2 to 9 participating nodes under 50% majority voting [3].

D. FedID Performance and Benchmark Validation

FedID utilizes XGBoost and FedAvg over 100 communication rounds. Without data sharing (S=0%), Non-IID traffic causes model divergence (cosine similarity drops to 0.19 for Blackhole). Sharing 10% of local samples elevates cosine similarity above 0.90 across all attacks (Fig. 6). At S=10%, FedID achieves 95.7% (BH), 82.4% (SF), 87.0% (DI), 82.6% (DR), 98.0% (HF), 100% (IV), and 87.4% (WP) accuracy, thoroughly outperforming CIDwL while incurring **zero communication bytes during testing** (Table III and Fig. 7).

TABLE III
FEDID DETECTION ACCURACY ACROSS SHARING RATIOS VS CIDWL [3]

Attack	FedID (0%)	FedID (5%)	FedID (10%)	CIDwL
BH	0.841	0.918	0.957	0.621
SF	0.641	0.741	0.824	0.550
DI	0.701	0.806	0.870	0.647
DR	0.640	0.741	0.826	0.548
HF	0.882	0.963	0.980	0.683
IV	1.000	1.000	1.000	0.998
WP	0.698	0.795	0.874	0.608

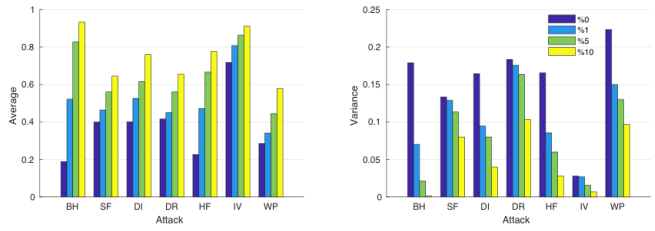


Fig. 6. Model divergence (mean and variance of cosine similarity) vs. data sharing ratio [3].

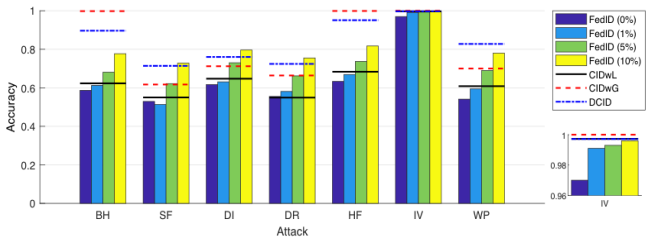


Fig. 7. Comparative testing accuracy of FedID vs. CIDwL, CIDwG, and DCID baselines [3].

On the real-world CICIoT2023 benchmark [13], FedID achieves **99.60% binary accuracy** and **98.80% multiclass accuracy** across 34 classes, outperforming FL-MLP (70.53%) and LSTM (93.13%) implementations (Table IV).

TABLE IV
BENCHMARK COMPARISON ON CICIOT2023 DATASET [3]

Study	Algorithm / Model	Binary Acc.	Multi Acc.
Abbas (2023) [11]	FL with MLP	99.00%	N.A.
Nkoro (2024)	Centralized MLP	99.10%	70.53%
Wang (2023)	Centralized LSTM	N.A.	93.13%*
FedID [3]	FL with XGBoost	99.60%	98.80%

VI. LIMITATIONS AND REMAINING RESEARCH

While FedID significantly advances the state-of-the-art, several open challenges remain in practice:

1) *Evasive Adversaries:* If an intelligent insider discovers stationary monitor locations, it can relocate its attacks to unmonitored subtrees.

2) *Training Overhead:* Increasing XGBoost tree depth scales parameter exchange (2^d parameters), creating training overhead.

3) *Byzantine Poisoning Vulnerability:* Standard FedAvg assumes benign collaborators. A compromised insider node participating in federated rounds can inject poisoned gradients or falsified tree splits, degrading global decision boundaries.

VII. PROPOSED NOVELTY FOR UPCOMING ITERATIONS

To address the aforementioned limitations, we formulate **Dynamic and Resilient FedID (DR-FedID)**, depicted in Fig. 8:

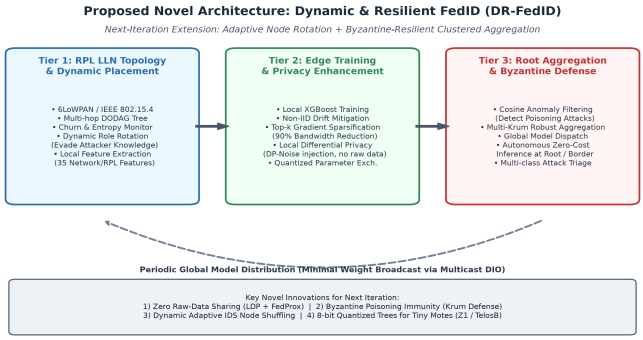


Fig. 8. Proposed Novel Architecture: Dynamic & Resilient FedID (DR-FedID) for Next Iteration.

1) **Dynamic Adaptive Monitor Shuffling (DA-FedID):** Collaborator nodes track routing churn and residual energy. When churn exceeds an adaptive entropy threshold, monitoring duties rotate dynamically among peer nodes, defeating adversary evasion.

2) **Byzantine-Resilient Multi-Krum Aggregation:** The root executes Multi-Krum and cosine anomaly filtering to reject weight updates that deviate by more than 2.5 standard deviations from median trajectories, neutralizing model poisoning attacks.

3) **Local Differential Privacy (LDP):** Replaces FedID's 10% raw sample

sharing by injecting calibrated Gaussian noise into top-k sparsified gradient sketches, providing formal (ε, δ)-differential privacy without raw telemetry exposure.

VIII. REPRODUCIBILITY AND CONCLUSION

All simulation source code, network configurations, result plotting scripts, and LaTeX documentation are openly available on GitHub at: https://github.com/mdilyas-27/cns_project_team_19. All experimental runs used deterministic random seeds and standardized Contiki 2.7 parameters to ensure complete reproducibility.

In conclusion, this project demonstrated that intrusion detection in RPL networks is fundamentally governed by topological placement. Centralized root monitoring fails against localized downstream attacks, whereas brute-force global feature streaming causes severe network saturation. Federated Intrusion Detection (FedID) successfully resolves this conflict, achieving up to 98.0% detection accuracy and 99.60% benchmark accuracy with zero runtime communication overhead. For upcoming iterations, our proposed DR-FedID framework will be prototyped and validated on the physical FIT IoT-LAB testbed, advancing resilient and energy-autonomous cybersecurity for the Internet of Things.

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